Optimization Methods in Machine Learning, CUB, Spring 2026

Theory 3. Constrained optimization and dual problems.

Danila Biktimirov

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1. Task 1

Solve $\min_{x \in \mathbb{R}^n} \{ c^T x : x^T A x \le 1 \}$, $c \ne 0$, $A \in S^n_{++}$.

Substitute $y = A^{1/2} x$:

$$\min_{\|y\| \le 1} \left(A^{-1/2} c \right)^T y.$$

By Cauchy–Schwarz,

$$\left(A^{-1/2} c \right)^T y \ge -\|A^{-1/2} c\| \, \|y\| \ge -\sqrt{c^T A^{-1} c},$$

with equality at $y^* = -A^{-1/2} c / \, \|A^{-1/2} c\|$. Hence

$$x^* = -\frac{A^{-1} c}{\sqrt{c^T A^{-1} c}}, \quad p^* = -\sqrt{c^T A^{-1} c}.$$

The Cauchy–Schwarz equality case is unique since $A^{-1/2} c \ne 0$.

2. Task 2

Solve $\min_{X \in S^n_{++}} \{ \operatorname{tr}(X^{-1}) : \langle A, X \rangle \le b \}$, $A \in S^n_{++}$, $b > 0$.

If $\langle A, X \rangle < b$, then tX is feasible for some $t > 1$ and $\operatorname{tr}\left((tX)^{-1}\right) = t^{-1} \operatorname{tr}(X^{-1}) < \operatorname{tr}(X^{-1})$, so the constraint is active.

Lagrangian:

$$L(X, \lambda) = \operatorname{tr}(X^{-1}) + \lambda(\langle A, X \rangle - b), \quad \lambda \ge 0.$$

Using $d \operatorname{tr}(X^{-1})[H] = -\operatorname{tr}(X^{-2} H)$, stationarity gives

$$-X^{-2} + \lambda A = 0 \Longrightarrow X^{-2} = \lambda A.$$

The positive definite square root is unique, so

$$X^{-1} = \sqrt{\lambda} A^{1/2}, \quad X = \lambda^{-1/2} A^{-1/2}.$$

Active constraint: $b = \operatorname{tr}(AX) = \lambda^{-1/2} \operatorname{tr}\left(A^{1/2}\right)$, hence $\sqrt{\lambda} = \operatorname{tr}\left(A^{1/2}\right) / b$ and

$$X^* = \frac{b}{\operatorname{tr}\left(A^{1/2}\right)} A^{-1/2}, \quad p^* = \frac{\left(\operatorname{tr}\left(A^{1/2}\right)\right)^2}{b}.$$

3. Task 3

Let $\lambda_1, \dots, \lambda_n > 0$ be the eigenvalues of $X \in S^n_{++}$. Since $X = X^T$,

$$\sum_{i=1}^{n} \|Xe_i\|^2 = \sum_{i=1}^{n} e_i^T X^2 e_i = \operatorname{tr}(X^2) = \sum_{i=1}^{n} \lambda_i^2.$$

The constraints $\|Xe_i\| \le 1$ give $\operatorname{tr}(X^2) \le n$. By AM–GM,

$$\det X = \prod \lambda_i = \left(\prod \lambda_i^2\right)^{1/2} \le \left(\frac{1}{n} \sum \lambda_i^2\right)^{n/2} \le 1.$$

I_n is feasible with $\det I_n = 1$, so it is optimal. Equality forces all $\lambda_i^2 = 1$, and $X \succ 0$ gives $\lambda_i = 1$, hence $X^* = I_n$ uniquely.

Hadamard. For arbitrary $X \in S_{++}^n$, set $d_i = \|Xe_i\| > 0$, $D = \operatorname{diag}(d_1, ..., d_n)$, $B = XD^{-1}$. Then $\|Be_i\| = 1$. Let $H = \left(B^T B\right)^{1/2} \in S_{++}^n$. For every i,

$$\|He_i\|^2 = e_i^T B^T B e_i = \|Be_i\|^2 = 1,$$

so H is feasible above and $\det H \le 1$. But $\det H = \sqrt{\det(B^T B)} = |\det B|$, so $|\det B| \le 1$, and

$$\det X = \det B \prod d_i \le \prod \|Xe_i\|.$$

4. Task 4

Solve $\min_{X \in S_{++}^n} \left\{\langle C^{-1}, X\rangle - \log\det X : a^T X a \le 1\right\}$, $C \in S_{++}^n$, $a \ne 0$.

The objective is strictly convex on S_{++}^n, the constraint is affine, and Slater holds (e.g. $X = \varepsilon I_n$ small). Hence KKT is necessary and sufficient, the solution is unique.

Lagrangian:

$$L(X, \lambda) = \langle C^{-1}, X\rangle - \log\det X + \lambda(a^T X a - 1), \quad \lambda \ge 0.$$

Stationarity:

$$C^{-1} - X^{-1} + \lambda a a^T = 0 \Longrightarrow X = \left(C^{-1} + \lambda a a^T\right)^{-1}.$$

Let $\gamma = a^T C a > 0$. By the Woodbury identity,

$$X = C - \frac{\lambda}{1 + \lambda\gamma} C a a^T C, \quad a^T X a = \gamma - \frac{\lambda\gamma^2}{1 + \lambda\gamma} = \frac{\gamma}{1 + \lambda\gamma}.$$

Complementary slackness: $\lambda(\gamma/(1 + \lambda\gamma) - 1) = 0$.

- If $\gamma \le 1$: $\lambda^* = 0$, so $X^* = C$.
- If $\gamma > 1$: constraint active, $\gamma/(1 + \lambda\gamma) = 1$ gives $\lambda^* = (\gamma - 1)/\gamma$.

Combined,

$$X^* = C - \frac{(\gamma - 1)_+}{\gamma^2} C a a^T C, \quad \gamma = a^T C a, \quad (t)_+ = \max\{t, 0\}.$$

5. Task 5

Primal: $\min_{x,s} \left\{\frac{1}{2} \|s - b\|^2 + \frac{\rho}{2} \|x\|^2 : s = Ax\right\}$, $\rho > 0$. Introduce $\nu \in \mathbb{R}^m$:

$$L(x, s, \nu) = \frac{1}{2} \|s - b\|^2 + \frac{\rho}{2} \|x\|^2 + \nu^T (s - Ax).$$

Minimization over s and x:

$$s-b+\nu=0 \Longrightarrow s=b-\nu, \quad \rho x-A^T\nu=0 \Longrightarrow x=\frac{1}{\rho}A^T\nu.$$

Substituting:

$$g(\nu)=b^T\nu-\frac{1}{2}\,\|\nu\|^2-\frac{1}{2\rho}\,\|A^T\nu\|^2.$$

Dual problem (unconstrained concave quadratic):

$$\max_{\nu\in\mathbb{R}^m} b^T\nu-\frac{1}{2}\,\|\nu\|^2-\frac{1}{2\rho}\,\|A^T\nu\|^2,$$

or equivalently $\min_\nu \frac{1}{2}\nu^T(I_m+(1/\rho)AA^T)\nu-b^T\nu$, with maximizer

$$\nu^*=\left(I_m+\frac{1}{\rho}AA^T\right)^{-1}b.$$

6. Task 6

Let $\Delta_{ik}=\mathbb{1}\{k\neq y_i\}$. The two primal constraints combine into

$$w_k^Tx_i+b_k-w_{y_i}^Tx_i-b_{y_i}+\Delta_{ik}-\xi_i\le 0, \quad \forall i,k,$$

since the $k=y_i$ case is exactly $\xi_i\ge 0$. With multipliers $\beta_{ik}\ge 0$,

$$L=\frac{1}{2}\sum_k\|w_k\|^2+C\sum_i\xi_i+\sum_{i,k}\beta_{ik}\left[w_k^Tx_i+b_k-w_{y_i}^Tx_i-b_{y_i}+\Delta_{ik}-\xi_i\right].$$

$\partial L/\partial\xi_i$: $C-\sum_k\beta_{ik}=0$.

Define

$$\alpha_{ik}=\begin{cases}-\beta_{ik} & k\neq y_i\\ C-\beta_{iy_i} & k=y_i\end{cases}$$

Then $\sum_k\alpha_{ik}=0$, $\alpha_{ik}\le 0$ for $k\neq y_i$, and $\alpha_{iy_i}=\sum_{k\neq y_i}\beta_{ik}\le\sum_k\beta_{ik}=C$.

$\partial L/\partial w_\ell$:

$$w_\ell+\sum_i\left(\beta_{i\ell}-\mathbb{1}\{y_i=\ell\}\sum_k\beta_{ik}\right)x_i=0 \Longrightarrow w_\ell=\sum_i\alpha_{i\ell}x_i.$$

Linear part of the dual:

$$\sum_{i,k}\beta_{ik}\Delta_{ik}=\sum_i\sum_{k\neq y_i}\beta_{ik}=\sum_i\alpha_{iy_i}.$$

Quadratic part:

$$-\frac{1}{2}\sum_\ell\|w_\ell\|^2=-\frac{1}{2}\sum_{i,j}x_i^Tx_j\sum_{k=1}^K\alpha_{ik}\alpha_{jk}.$$

Hence the dual

$$\max_{\alpha} \sum_{i=1}^{N} \alpha_{iy_i} - \frac{1}{2} \sum_{i,j=1}^{N} x_i^T x_j \sum_{k=1}^{K} \alpha_{ik}\alpha_{jk}$$

subject to

$$\sum_{k=1}^{K} \alpha_{ik} = 0, \quad \alpha_{ik} \leq 0 \quad (k \neq y_i), \quad \alpha_{iy_i} \leq C, \quad \forall i.$$

Since $\sum_k \alpha_{ik} = 0$, the linear term satisfies $\sum_i \alpha_{iy_i} = -\sum_i \sum_{k \neq y_i} \alpha_{ik}$, which matches the form $\sum_{i,k} \alpha_{ik}$ (modulo sign conventions) of the stated objective.